2023-2026

PORTFOLIO

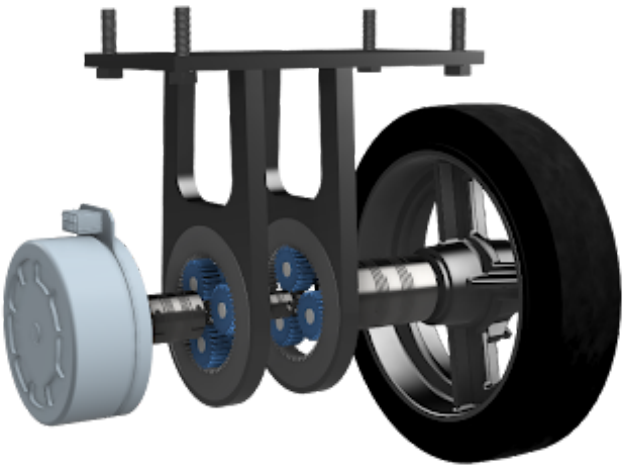
Mechanical
Engineering

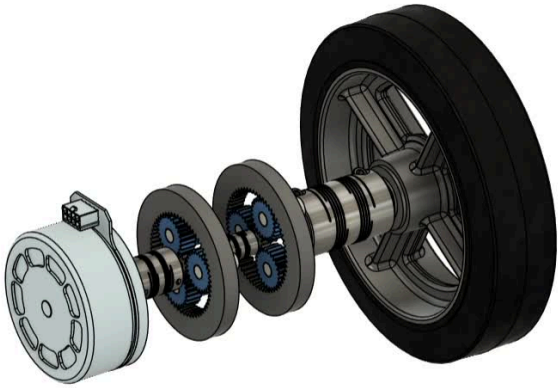
Lena Rosario

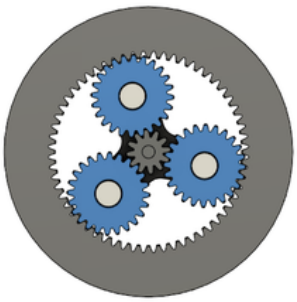
Planetary Gear Drive

This project was to design a rear wheel drive system for a hospital delivery robot. The task was to design one rear wheel and the components needed to move it.

The parameters for this design include a specific payload, required torque, along with dimension limitations and coefficients of static friction and rolling resistance.



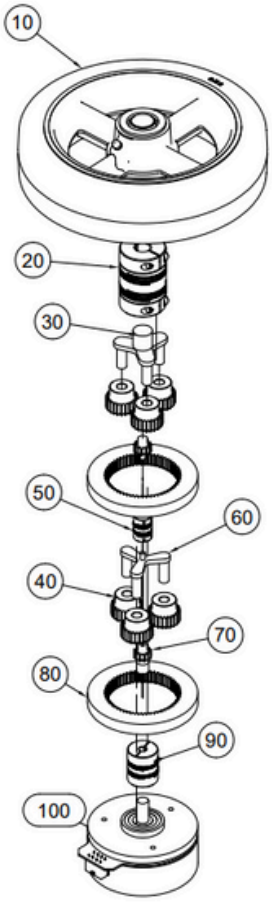




This gear drive was created using Fusion 360. I sourced gears from McMaster Carr and estimated a cost to produce this drive.

Parts List

Item	Qty	Part Number	Part Name
10	1	8368T31	Rubber Wheel
20	1	2464K46	Shaft Coupling Large
30	1	Custom	Carrier 2
40	6	2664N542	Planet Gear
50	1	2464K11	Shaft Coupling Small
60	1	Custom	Carrier 1
70	2	2662N332	Sun Gear
80	2	2696N27	Ring Gear
90	1	2464K22	Shaft Coupling Medium
100	1	MEEN 15	Motor



Plantary Gear Wheel Drive Assembly

Lena Rosario

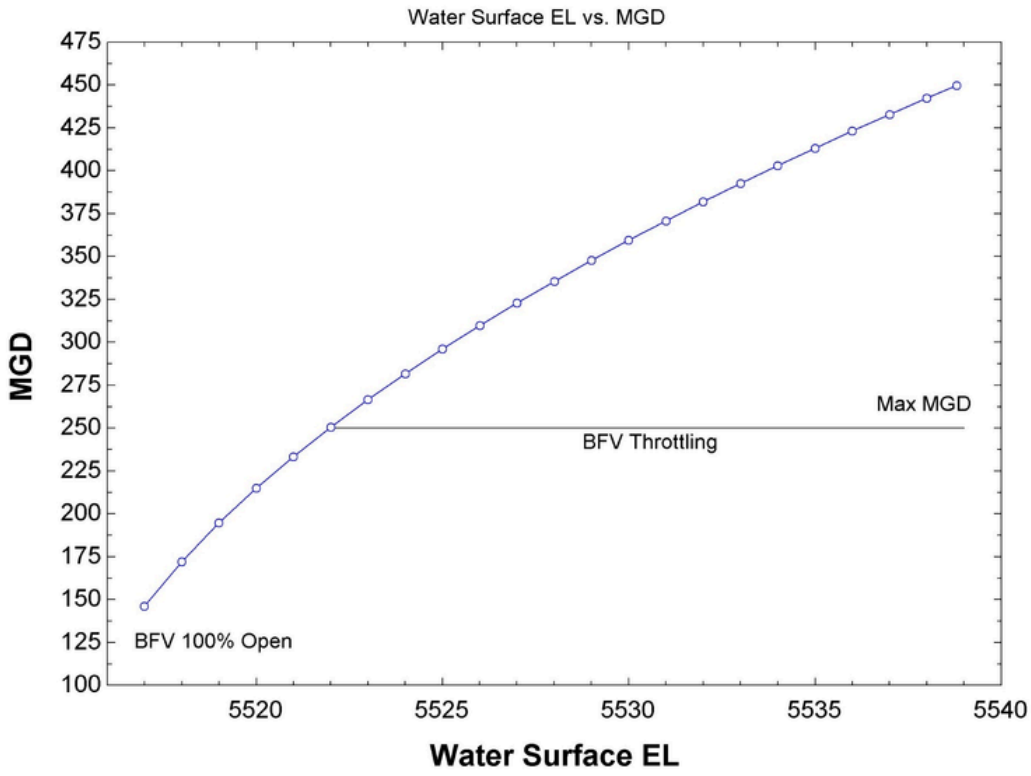
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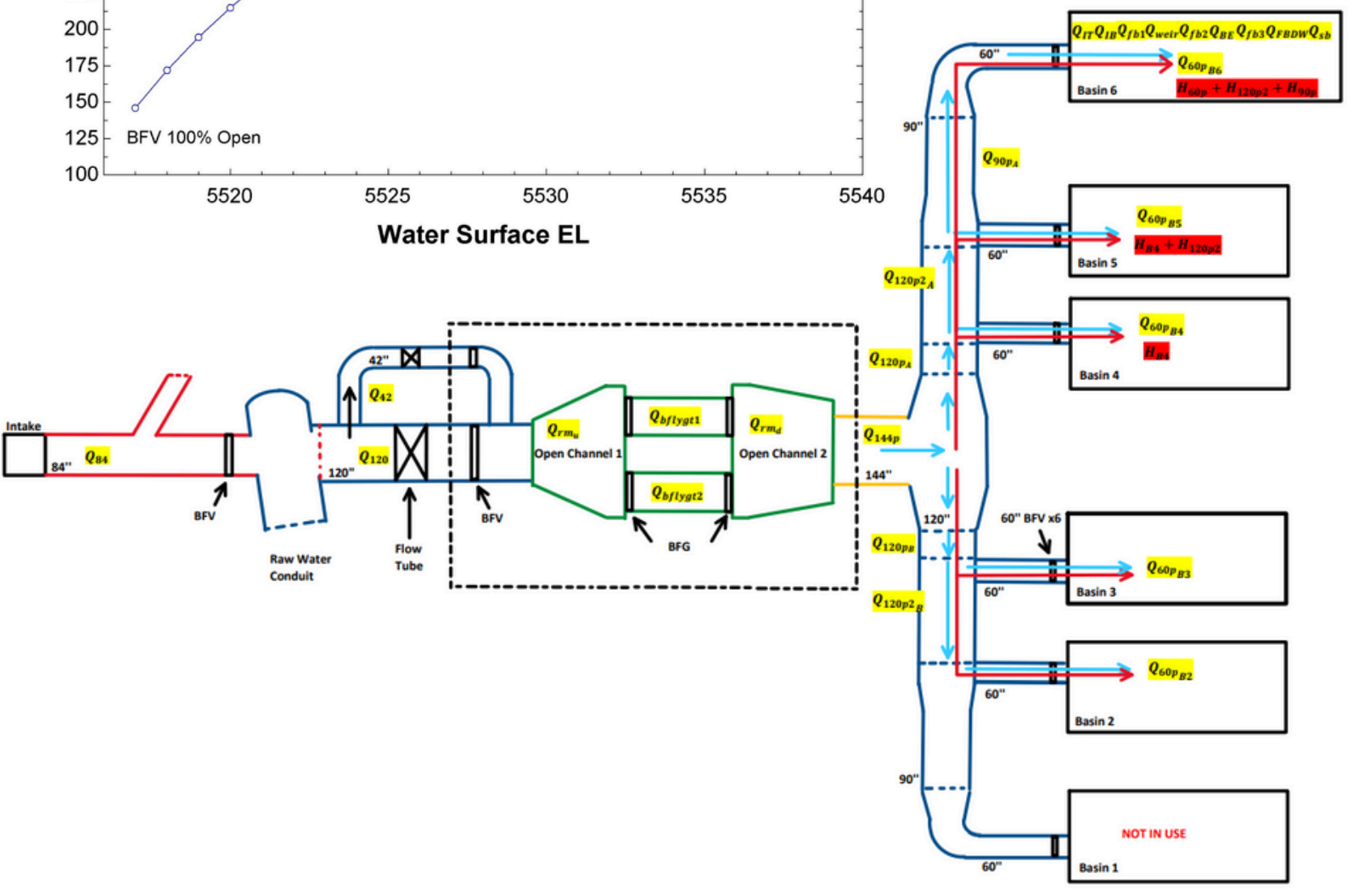
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Hydraulic Model EES

During my internship at Denver Water I created a model to predict the number of days the water in a specific reservoir could be relied on to meet water demand needs during peak usage season.

Using construction drawings, I modeled the hydraulic losses through the outlet works piping using Engineering Equation Solver. I modeled the head loss through each element of the piping geometry to ultimately determine the flow rate of water entering the treatment plant as the water level in the reservoir goes down.

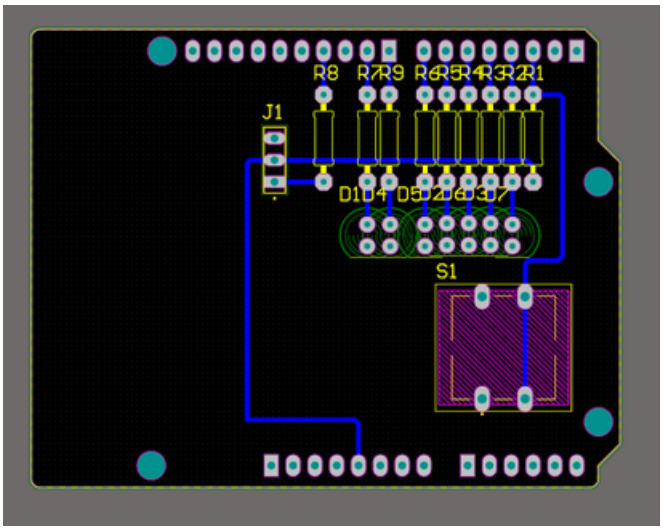


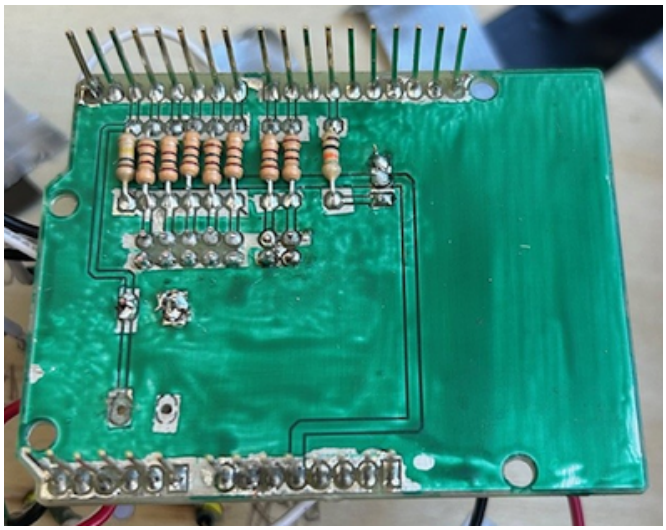


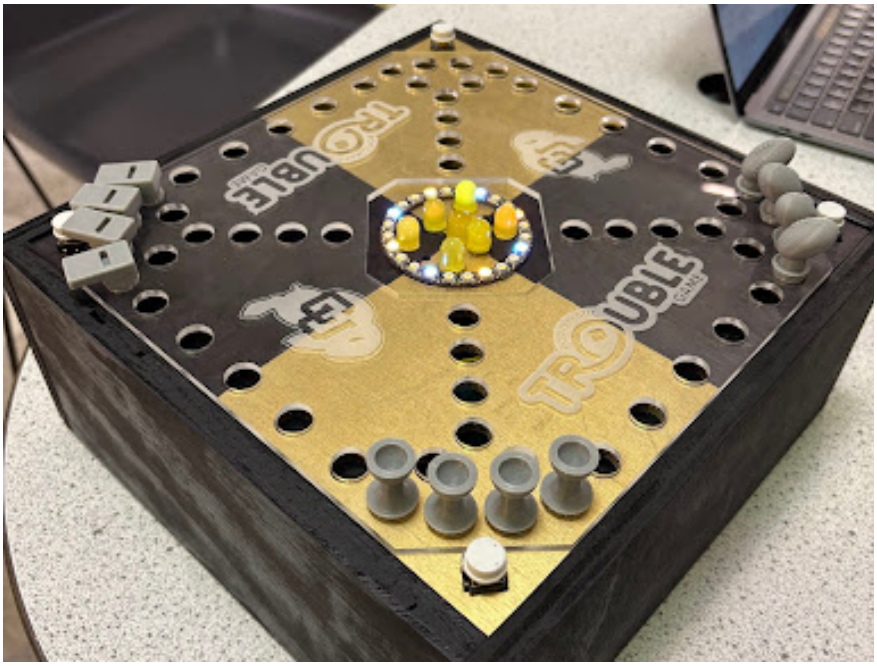
Electronic Trouble! Game

This project was to re-create a board game of our choice but add an electronic component to the game. We chose the board game Trouble! and chose to create an electronic dice rolling mechanism.

We designed the needed circuit, created it in Altium and printed a custom PCB board. I soldered all the necessary resistors and LEDs on the board and did work troubleshooting the system during assembly.







Using Arduino, an LED light ring and 6 LED bulbs we could simulate a dice rolling function when a player pushes the button in each corner.

MSR Stove Reverse Engineering

This project was to reverse engineer an object of our choice over the semester and create a detailed CAD model of the object. I chose my MSR backpacking stove and created the model using SolidWorks.





I limited the joints of the prongs of the stove to move and fold down the same way as the actual stove function

